

Hayk Tserunyan

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TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML, CSS
Frontend: React, Tailwind CSS, Bootstrap
Backend: Spring Boot, Node.js, Express, Flask, FastAPI
Databases: MongoDB, PostgreSQL, MySQL, TimescaleDB
DevOps/Tools: Git, GitHub, Docker, Postman, CI/CD (GitHub Actions)
Data & Automation: Pandas, NumPy, Matplotlib, Scikit-learn, Web Scraping

EXPERIENCE

Backend Development Intern

Jan 2024 – Aug 2024

Cast & Crew

Burbank, CA

- Built and maintained backend features in Node.js/Express supporting core digital payroll workflows
- Optimized MongoDB indexing + queries, improving high-traffic API performance by 40%
- Participated in code reviews and collaborated in Agile sprints (standups, planning, peer reviews)
- Implemented authentication, logging, and rate-limiting middleware to improve security and reliability
- Maintained CI/CD pipelines using GitHub Actions for automated testing and deployments

API Team Programmer

March 2023 – June 2023

Zotbins

Irvine, CA

- Developed the foundation for a CI pipeline, reducing manual testing effort by 30% and enabling safer deployments
- Improved database throughput by integrating TimescaleDB time-series optimizations, resulting in up to 5x faster inserts and 3x faster deletes
- Refactored backend components to reduce technical debt, decreasing code complexity by approximately 20%
- Created internal documentation improving onboarding efficiency by 40%

PROJECTS

AI License Plate Detection System | *Java, Spring Boot, MongoDB, Python, OpenCV*

- Built a full-stack traffic monitoring system using OpenCV-based OCR for license plate detection
- Developed a modular Spring Boot backend for violation tracking, citation issuance, and audit logging
- Integrated a Python OCR microservice for image processing and structured data serialization
- Modeled system data using MongoDB to support dynamic schemas for images and violation records
- Built an administrative dashboard using Spring MVC and Thymeleaf to visualize real-time detections and system logs

Document Management System | *Spring Boot, React, PostgreSQL, Python OCR, Docker*

- Built a full-stack document processing platform supporting PDF/image upload and automated text extraction
- Developed a Spring Boot backend with JWT-secured REST APIs for file upload, metadata tagging, and full-text search
- Implemented OCR and classification pipelines using FastAPI, Tesseract, and OpenCV
- Containerized the backend, OCR service, and database with Docker Compose for multi-service orchestration
- Designed a React dashboard for document preview, search, filtering, and user management

Python Automation & IT Toolkit | *Python, Playwright, Selenium, Flask, FastAPI, BeautifulSoup, Pandas*

- Built automation scripts using Python, Playwright, and Selenium to scrape product data, track prices, and collect research datasets across multiple websites
- Developed IT utilities for file management, including batch renaming, directory cleanup, and log parsing tools to reduce repetitive manual work
- Implemented system monitoring scripts for CPU, RAM, disk usage, and process activity, with optional email or webhook alerts for threshold violations
- Designed lightweight Flask/FastAPI microservices to expose automation jobs as REST APIs for remote execution and scheduling

EDUCATION

California State University, Northridge

Bachelors in Computer Science

Northridge, CA

Aug. 2021 - May 2025